

PROGRAM 8 :

Step 1: Write the Node.js Application

```
rv24mc028_chethanacs@Elite:~/DEVops/prog8$ cat server.js
const http = require("http");

const port = 3000;

const server = http.createServer((req, res) => {
  res.end("Hello from Node.js running inside Kubernetes!");
});

server.listen(port, () => {
  console.log(`Server running at port ${port}`);
});
```

Step 2: Create a ConfigMap for the Node.js Code

```
rv24mc028_chethanacs@Elite:~/DEVops/prog8$ cat nodejs-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nodejs-pod
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      ports:
        - containerPort: 3000
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

Step 3: Create the Pod YAML Using Node.js

```
rv24mc028_chethanacs@Elite:~/DEVops/prog8$ cat nodejs-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nodejs-pod
  labels:
    app: nodejs
spec:
  containers:
  - name: nodejs
    image: node:18
    command: ["node", "/usr/src/app/server.js"]
    ports:
    - containerPort: 3000
    volumeMounts:
    - name: app-code
      mountPath: /usr/src/app
  volumes:
  - name: app-code
    configMap:
      name: nodejs-app-config
```

Step 4: Apply the YAML Files

```
rv24mc028_chethanacs@Elite:~/DEVops/prog8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config created
rv24mc028_chethanacs@Elite:~/DEVops/prog8$ kubectl apply -f nodejs-pod.yaml
pod/nodejs-pod created
```

Step 5: Check Pod Status

```
rv24mc028_chethanacs@Elite:~/DEVops/prog8$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
backup-job-29445715-s5bxd	0/1	Completed	0	3m2s
my-cron-29445715-fqw92	0/1	Completed	0	3m2s
my-cron-29445716-jkgd7	0/1	Completed	0	2m2s
my-cron-29445717-rkbl7	0/1	Completed	0	62s
my-cron-29445718-nszdl	0/1	ContainerCreating	0	2s
nodejs-pod	0/1	ContainerCreating	0	9s


```
rv24mc028_chethanacs@Elite:~/DEVops/prog8$ kubectl get pods
```

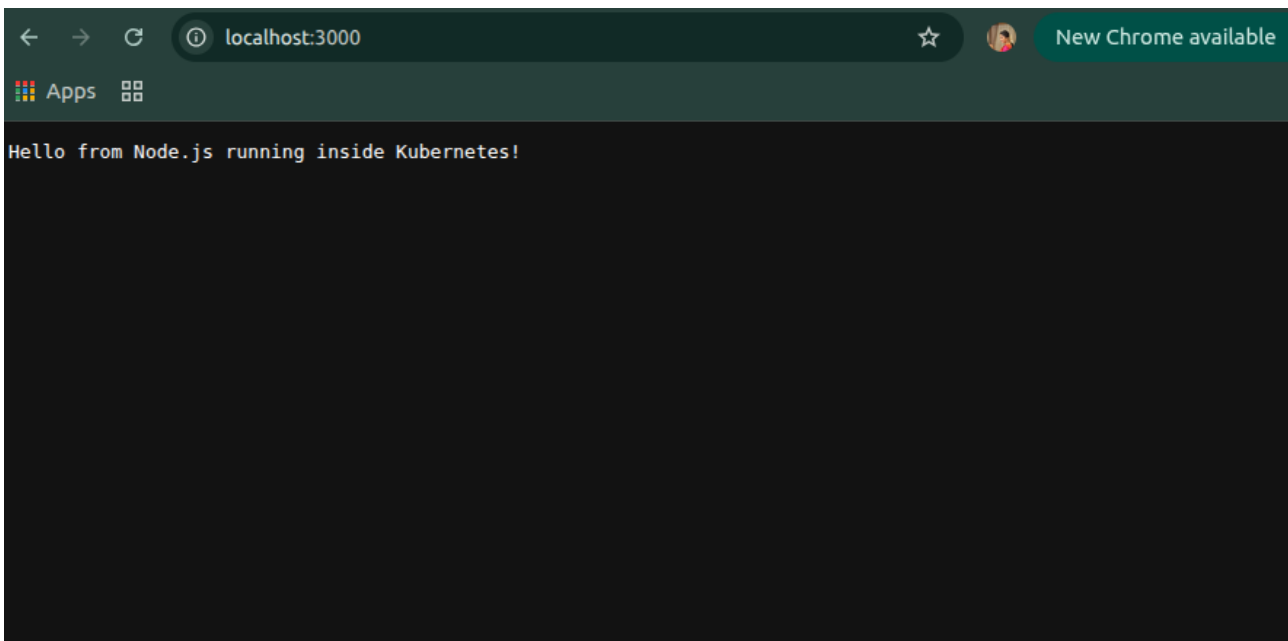
NAME	READY	STATUS	RESTARTS	AGE
backup-job-29445715-s5bxd	0/1	Completed	0	12m
backup-job-29445720-4fpxz	0/1	Completed	0	7m12s
backup-job-29445725-6w9td	0/1	Completed	0	2m12s
my-cron-29445725-bmcg9	0/1	Completed	0	2m12s
my-cron-29445726-n7lz2	0/1	Completed	0	72s
my-cron-29445727-zq7pb	0/1	Completed	0	12s
nodejs-pod	1/1	Running	0	9m19s

Step 6: Access the Node.js Application

```
rv24mc028_chethanacs@Elite:~/DEVops/prog8$ kubectl port-forward pod/nodejs-pod 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
Handling connection for 3000
```

Step 7: Verify the Output

Output:



Step 8: Delete All Pods ,Services and Deployments

```
^Crv24mc028_chethanacs@Elite:~/DEVops/prog8$ kubectl delete pods --all
pod "backup-job-29445730-gzf9h" deleted from default namespace
pod "backup-job-29445735-f57sq" deleted from default namespace
pod "backup-job-29445740-rpn52" deleted from default namespace
pod "my-cron-29445742-nznw4" deleted from default namespace
pod "my-cron-29445743-ztdlm" deleted from default namespace
pod "my-cron-29445744-gk6b6" deleted from default namespace
pod "nodejs-pod" deleted from default namespace
rv24mc028_chethanacs@Elite:~/DEVops/prog8$ kubectl delete svc --all
service "kubernetes" deleted from default namespace
rv24mc028_chethanacs@Elite:~/DEVops/prog8$ kubectl delete deployments --all
```